

Windows Server 2025 Environment — Implementation Plan (2026 Design)

Derived exclusively from WSC2026_TP39_topology_pre.pdf | Parameterised build framework

23 September 2026

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Document Control & Basis of Plan

Source and scope

This plan is derived **exclusively** from WSC2026_TP39_topology_pre.pdf. No values have been carried over from the 2024 Module B document or from any other source.

Critical limitation

WSC2026_TP39_topology_pre.pdf is a **pre-release** document. Every FQDN, IP address, subnet, domain name and per-host service label on the Module B Windows diagram is printed as <redacted>.

Consequence: a build-ready plan cannot be produced from this source alone. What follows is a **parameterised framework** — the architecture, phasing, dependency logic, cmdlets and validation tests are complete and correct; the values are placeholders.

What the 2026 source actually establishes

Confirmed element	Value
Sites	Shanghai, Beijing
Zones	Internet, DMZ, INTERNAL, CLIENT
Server OS baseline	Windows Server 2025 — Core and GUI (Desktop Experience) editions both in scope
Client OS baseline	Windows 11
Client tooling	RSAT, WSL (Debian)

Confirmed element	Value
DMZ composition	Two Core servers per site

What the 2026 source does not establish

All of the following are [TO BE CONFIRMED] and must be supplied before day one:

- Every hostname and FQDN
- Every IP address, subnet mask and VLAN ID
- Forest root domain name, child/subordinate domain names, NetBIOS names
- Forest and domain functional levels
- AD site names, site links and subnet-to-site mappings
- Which Windows Server role runs on which host
- DNS zone names, forwarders and delegation points
- DHCP scope ranges, exclusions, lease durations and options
- Storage platform, hypervisor product, VM specifications
- Backup, antivirus and monitoring products
- Certificate authority naming, hierarchy depth and template requirements
- Edge firewall product and rule set between Internet, DMZ, INTERNAL and CLIENT

How to use this document

Section 4.0 defines a single variable block. Populate it once from the released (unredacted) topology, and every PowerShell block in Sections 4 to 6 becomes executable without further edit.

1. Environment Summary

1.1 Zone architecture

The 2026 topology defines four zones. Their security posture and permitted traffic direction are inferred from standard zone semantics; **the actual firewall rule set is not in the source** and is [TO BE CONFIRMED].

Zone	Trust level	Contains (per source)	Inbound from	Outbound to
Internet	Untrusted	External clients and services	—	DMZ only
DMZ	Semi-trusted	Two Core servers per site	Internet (published services only)	INTERNAL (brokered, specific ports only)
INTERNAL	Trusted	Server estate [TO BE	DMZ (restricted), CLIENT	DMZ, CLIENT

Zone	Trust level	Contains (per source)	Inbound from	Outbound to
		CONFIRMED]		
CLIENT	Trusted (user)	Windows 11 endpoints with RSAT and WSL (Debian)	INTERNAL	INTERNAL

Design note: the presence of a discrete DMZ zone is the single most significant architectural feature of the 2026 design. It mandates a firewalled boundary, a published-services model, and almost certainly a reverse proxy or application gateway tier. None of the mechanism is specified in the source.

1.2 Site design

Site	Zones present	Role	Inter-site link
Shanghai	Internet, DMZ, INTERNAL, CLIENT	[TO BE CONFIRMED] — hub or peer	[TO BE CONFIRMED] — WAN, VPN or MPLS not specified
Beijing	Internet, DMZ, INTERNAL, CLIENT	[TO BE CONFIRMED] — hub or peer	[TO BE CONFIRMED]

Both sites carry a full four-zone stack, which implies a **symmetric, dual-site design** rather than a hub-and-spoke branch model. Whether this represents active/active, active/passive or independent-per-site operation is [TO BE CONFIRMED].

1.3 Node inventory

Hostnames are **not** in the source. The identifiers below are positional placeholders describing *where a node sits*, not names to be used. Replace all of them.

Placeholder ID	Site	Zone	Hostname	Role	OS / Edition	IP / VLAN	Specs	Dependencies
SH-DMZ-01	Shanghai	DMZ	[TO BE CONFIRMED]	[TO BE CONFIRMED]	Windows Server 2025 Core	[TO BE CONFIRMED]	[TO BE CONFIRMED]	Edge firewall; INTERNAL brokered services
SH-DMZ-02	Shanghai	DMZ	[TO BE CONFIRMED]	[TO BE CONFIRMED]	Windows Server 2025 Core	[TO BE CONFIRMED]	[TO BE CONFIRMED]	Edge firewall; SH-DMZ-01
BJ-DMZ-01	Beijing	DMZ	[TO BE CONFIRMED]	[TO BE CONFIRMED]	Windows Server 2025 Core	[TO BE CONFIRMED]	[TO BE CONFIRMED]	Edge firewall; INTERNAL brokered services
BJ-DMZ-02	Beijing	DMZ	[TO BE CONFIRMED]	[TO BE CONFIRMED]	Windows Server 2025 Core	[TO BE CONFIRMED]	[TO BE CONFIRMED]	Edge firewall; BJ-DMZ-01

Placeholder ID	Site	Zone	Hostname	Role	OS / Edition	IP / VLAN	Specs	Dependencies
SH-INT-nn	Shanghai	INTERNAL	[TO BE CONFIRMED]	[TO BE CONFIRMED]	Windows Server 2025 Core / GUI [TBC]	[TO BE CONFIRMED]	[TO BE CONFIRMED]	Site DC; DNS
BJ-INT-nn	Beijing	INTERNAL	[TO BE CONFIRMED]	[TO BE CONFIRMED]	Windows Server 2025 Core / GUI [TBC]	[TO BE CONFIRMED]	[TO BE CONFIRMED]	Site DC; DNS
SH-CLI-nn	Shanghai	CLIENT	[TO BE CONFIRMED]	User endpoint	Windows 11 + RSAT + WSL (Debian)	[TO BE CONFIRMED]	[TO BE CONFIRMED]	Site DHCP, DNS, DC
BJ-CLI-nn	Beijing	CLIENT	[TO BE CONFIRMED]	User endpoint	Windows 11 + RSAT + WSL (Debian)	[TO BE CONFIRMED]	[TO BE CONFIRMED]	Site DHCP, DNS, DC

INTERNAL zone node count is not stated in the source. The table shows generic rows; the actual inventory is [TO BE CONFIRMED].

1.4 Addressing, domain and forest design

Element	Value
Forest root domain	[TO BE CONFIRMED]
Child / subordinate domains	[TO BE CONFIRMED] — the source does not indicate whether the design is single-domain or multi-domain
NetBIOS names	[TO BE CONFIRMED]
Forest / domain functional level	[TO BE CONFIRMED] — recommend WinServer2025 if all DCs are 2025
AD sites	[TO BE CONFIRMED] — recommend Shanghai and Beijing to mirror the physical topology
Site link cost / schedule	[TO BE CONFIRMED]
Subnets and masks	[TO BE CONFIRMED] — one per zone per site, eight minimum
VLAN IDs	[TO BE CONFIRMED] — zone separation implies tagging, but no IDs are given
DNS zones and forwarders	[TO BE CONFIRMED]
DHCP scopes	[TO BE CONFIRMED] — CLIENT zones in both sites will require scopes

Element	Value
Storage / virtualization	[TO BE CONFIRMED] — not depicted
Edge / internet boundary	Present in topology as the Internet zone; device, product and rules [TO BE CONFIRMED]

1.5 Minimum subnet plan (structure only)

Eight subnets are the architectural minimum implied by four zones across two sites. Values are [TO BE CONFIRMED].

Site	Zone	Subnet	VLAN	Gateway	DHCP
Shanghai	DMZ	[TBC]	[TBC]	[TBC]	Static
Shanghai	INTERNAL	[TBC]	[TBC]	[TBC]	Static
Shanghai	CLIENT	[TBC]	[TBC]	[TBC]	Yes
Shanghai	Internet / edge transit	[TBC]	[TBC]	[TBC]	Static
Beijing	DMZ	[TBC]	[TBC]	[TBC]	Static
Beijing	INTERNAL	[TBC]	[TBC]	[TBC]	Static
Beijing	CLIENT	[TBC]	[TBC]	[TBC]	Yes
Beijing	Internet / edge transit	[TBC]	[TBC]	[TBC]	Static

2. Services and Roles Inventory

The source **does not assign services to hosts** — every service label on the Module B diagram is redacted. The table below lists the roles structurally required by a four-zone, two-site Windows Server 2025 forest. Host assignment is [TO BE CONFIRMED] in every row.

#	Role / feature / software	Host	Why required	Prerequisites	Edition / licence	Ports & protocols
1	AD DS (AD-Domain-Services)	[TBC] — INTERNAL, both sites	A forest is implied by the domain-joined CLIENT and INTERNAL zones	Static IP; final hostname; time sync	Server 2025 Std or Datacenter	TCP 88, 135, 389, 445, 464, 636, 3268/3269, 49152–65535; UDP 88, 123, 389, 464
2	DNS Server	[TBC] — INTERNAL,	Domain join and AD	AD DS (if AD-	Included	TCP/UDP 53

#	Role / feature / software	Host	Why required	Prerequisites	Edition / licence	Ports & protocols
		both sites	locator records are impossible without it	integrated)		
3	DHCP Server	[TBC] — INTERNAL or per-site	CLIENT zones require dynamic addressing	Domain join; Add-DhcpServerInDC	Included	UDP 67/68; TCP 647 (failover)
4	DHCP Relay Agent (RRAS)	[TBC] — CLIENT gateway	Required if the DHCP server is not on the CLIENT subnet. Confirm placement	RRAS	Included	UDP 67
5	AD CS — Certification Authority	[TBC] — INTERNAL, never DMZ	DMZ-published services will need server certificates; LDAPS needs a CA	Domain join; Enterprise Admin	Std supports Enterprise CA	TCP 135 + dynamic RPC; TCP 80 (CRL/AIA)
6	AD CS — Web Enrolment / CES / CEP	[TBC]	Enrolment for non-domain or DMZ hosts	Issuing CA online	Included	TCP 443
7	IIS (Web-Server)	[TBC] — DMZ Core servers	The DMZ zone exists to publish services; IIS is the probable vehicle	—	Included	TCP 80, 443
8	ARR + URL Rewrite	[TBC] — DMZ	Reverse proxy from Internet to INTERNAL origins without direct exposure	IIS; MSI packages	Free Microsoft add-ons	TCP 80, 443
9	Remote Access / RRAS	[TBC] — zone gateways	Inter-zone and inter-site routing	Two or more NICs; IP forwarding	Included	IP forwarding; UDP 500/4500
10	Site-to-site VPN (IKEv2)	[TBC] — Shanghai and Beijing edge	Inter-site AD replication path. Confirm whether the link is VPN, MPLS or private WAN	RRAS routing first	Included	UDP 500, 4500; ESP/50
11	File & Storage	[TBC] — INTERNAL	Implied by any user-	Domain join	Included	TCP 139, 445

#	Role / feature / software	Host	Why required	Prerequisites	Edition / licence	Ports & protocols
	Services		data requirement			
12	FSRM	[TBC] — INTERNAL	Quota and file-screen enforcement if required	FS-FileServer	Included	n/a (local)
13	DFS Namespaces / Replication	[TBC] — INTERNAL, both sites	Two-site design makes namespace abstraction and cross-site replication likely	DFS-N; domain join	Included	TCP 135, 389, 445 + dynamic RPC
14	RDS (Session Host, Broker, Web Access, Licensing)	[TBC]	Only if remote application or desktop publishing is in scope	Domain join; RD Licensing	RDS CALs required beyond 120-day grace	TCP/UDP 3389; TCP 443, 5504, 5985
15	Windows Admin Center	[TBC] — INTERNAL management host	Strongly recommended: four of the servers are Core with no local GUI	—	Free	TCP 443
16	RSAT	CLIENT endpoints	Explicitly specified in the 2026 topology	Windows 11	Included	—
17	WSL with Debian	CLIENT endpoints	Explicitly specified in the 2026 topology	Windows 11; virtualization enabled in firmware	Included / open source	Host-local; TCP 5985/5986 if used for WinRM automation
18	Edge firewall	Internet-to-DMZ-to-INTERNAL boundary	Mandatory given a discrete DMZ zone	—	[TO BE CONFIRMED]	[TO BE CONFIRMED]
19	Hypervisor	[TBC]	Platform not depicted in the source	—	[TO BE CONFIRMED]	—
20	Backup	[TBC]	Absent from the source. Mandatory before production	—	[TO BE CONFIRMED]	[TO BE CONFIRMED]

#	Role / feature / software	Host	Why required	Prerequisites	Edition / licence	Ports & protocols
21	Antivirus / EDR	All nodes	Absent from the source. Mandatory, especially for DMZ hosts	—	[TO BE CONFIRMED]	[TO BE CONFIRMED]
22	Monitoring	All nodes	Absent from the source	—	[TO BE CONFIRMED]	[TO BE CONFIRMED]

3. Build Order & Dependency Map

The sequence below is valid regardless of which values populate the placeholders, because it is driven by hard technical dependencies rather than by specific hosts.

Phase	Action	Why it must precede the next step	Parallel?
0	Platform: hypervisor, VM provisioning, virtual switches and VLANs for all eight zone/site segments, ISO and media staging	No Layer 3 work is possible without Layer 1 and 2	—
1	Edge firewall and inter-zone routing: Internet to DMZ to INTERNAL to CLIENT rule set, both sites	Zone traversal must work before any cross-zone dependency (domain join, replication, enrolment) can be tested	Yes — Shanghai and Beijing in parallel
2	Inter-site transport: VPN, MPLS or private WAN between Shanghai and Beijing, plus routing	Must precede any second-site DC promotion — promotion requires LDAP, Kerberos and RPC to the existing forest	—
3	First DC in the primary site: forest root, DNS installed with it	DNS must exist before any domain join. The entire estate depends on the _msdcs SRV records	—
4	DNS design: forward zones, reverse zones for all eight subnets, forwarders, conditional forwarders	Reverse resolution and forwarding must be correct before clients and member servers are introduced	—
5	AD Sites & Services: define Shanghai and Beijing sites, associate subnets, configure site link	Must precede the second-site DC — otherwise it lands in the default site and clients authenticate across the WAN	—

Phase	Action	Why it must precede the next step	Parallel?
6	Second DC in the primary site	Removes the single-DC failure window before the estate grows	Yes — with Phase 7
7	DC in the secondary site	Needs Phases 2, 3 and 5. Provides local authentication for that site's CLIENT zone	Yes — with Phase 6
8	DHCP: scopes for both CLIENT zones, relay agents on CLIENT gateways if the server is off-subnet	Authorisation in AD requires a live domain; relay must exist before clients can lease	—
9	PKI: AD CS hierarchy, INTERNAL zone only	PKI before LDAPS, before HTTPS bindings, before any certificate-authenticated VPN. Root distribution via Group Policy takes a replication cycle to land — build in the lead time	—
10	Certificate templates published, enrolment permissions delegated, autoenrolment GPO linked	Templates must be issuable before any host requests a certificate in Phase 11 or 12	—
11	INTERNAL member servers: domain join, role installation, application configuration	Roles that request certificates depend on Phase 10	Yes — across hosts
12	DMZ Core servers, both sites	Built last among servers and deliberately so: the DMZ is the exposed tier, so its origins in INTERNAL must be healthy and its certificates issuable before it is published to the Internet	Yes — both sites
13	AD DS structure: OUs, groups, service accounts, delegation	Group objects must exist before NTFS ACLs and GPO security filtering reference them	Groups can start in parallel from Phase 6
14	Group Policy: create, configure, link, verify precedence	Link targets are the Phase 13 OUs	—
15	File services, shares, NTFS permissions, DFS namespace and replication	References Phase 13 groups	—
16	CLIENT endpoints: Windows 11, domain join, RSAT, WSL with Debian	Need DHCP, DNS, DCs and GPOs live	Yes — across endpoints

Phase	Action	Why it must precede the next step	Parallel?
17	Publication: expose DMZ services through the edge firewall to the Internet zone	Last step by design. Nothing is exposed until the full internal chain is validated	—
18	Backup, antivirus/EDR and monitoring agents across the estate	Must be operational before handover, regardless of their absence from the source	Can begin at Phase 11

3.1 Dependency rules that must not be violated

1. **DNS before domain promotion** — a DC cannot be promoted against a forest it cannot resolve.
2. **Inter-site transport before remote-site promotion** — replication traffic has no path otherwise.
3. **AD Sites defined before the remote DC** — otherwise clients authenticate over the WAN and replication topology is wrong.
4. **PKI before LDAPS, HTTPS and certificate-based VPN** — certificates cannot be issued by a CA that does not exist, and root trust needs a replication cycle to distribute.
5. **AD groups before ACLs and GPO security filtering** — both reference SIDs that must already exist.
6. **INTERNAL healthy before DMZ published** — never expose a tier whose origin is unvalidated.

4. Detailed Installation Procedure

4.0 Variable block — populate this first

Every subsequent code block consumes these variables. Fill this in once from the released topology.

```
# =====
# WSC2026 TP39 - BUILD VARIABLES
# Populate every value below from the released (unredacted) topology
# before running any subsequent block in this document.
# =====

# --- Forest and domain -----
$ForestRoot      = "[TO BE CONFIRMED]" # e.g. contoso.local
$NetBiosName      = "[TO BE CONFIRMED]"
$ForestMode       = "[TO BE CONFIRMED]" # WinThreshold or Win2025
$DomainMode       = "[TO BE CONFIRMED]"
$DsrmPassword     = Read-Host -AsSecureString "DSRM password"

# --- Sites -----
$SitePrimary      = "Shanghai"
$SiteSecondary    = "Beijing"

# --- Subnets (one per zone per site) -----
$SH_DMZ_Subnet    = "[TO BE CONFIRMED]" # e.g. 10.x.x.0/24
$SH_INT_Subnet    = "[TO BE CONFIRMED]"
$SH_CLI_Subnet    = "[TO BE CONFIRMED]"
```

```

$SH_EDGE_Subnet = "[TO BE CONFIRMED]"
$BJ_DMZ_Subnet  = "[TO BE CONFIRMED]"
$BJ_INT_Subnet  = "[TO BE CONFIRMED]"
$BJ_CLI_Subnet  = "[TO BE CONFIRMED]"
$BJ_EDGE_Subnet = "[TO BE CONFIRMED]"

# --- VLAN IDs -----
$VlanDmz      = "[TO BE CONFIRMED]"
$VlanInternal = "[TO BE CONFIRMED]"
$VlanClient   = "[TO BE CONFIRMED]"

# --- Hosts -----
$DC1Name      = "[TO BE CONFIRMED]"
$DC1IP        = "[TO BE CONFIRMED]"
$DC1Gateway   = "[TO BE CONFIRMED]"
$DC2Name      = "[TO BE CONFIRMED]"
$DC2IP        = "[TO BE CONFIRMED]"
$DC3Name      = "[TO BE CONFIRMED]" # secondary site DC
$DC3IP        = "[TO BE CONFIRMED]"
$CAName       = "[TO BE CONFIRMED]"
$CAIP         = "[TO BE CONFIRMED]"
$CACommonName = "[TO BE CONFIRMED]"

# --- DHCP -----
$SH_ScopeStart = "[TO BE CONFIRMED]"
$SH_ScopeEnd   = "[TO BE CONFIRMED]"
$SH_ScopeMask  = "[TO BE CONFIRMED]"
$SH_ScopeRouter = "[TO BE CONFIRMED]"
$BJ_ScopeStart = "[TO BE CONFIRMED]"
$BJ_ScopeEnd   = "[TO BE CONFIRMED]"
$BJ_ScopeMask  = "[TO BE CONFIRMED]"
$BJ_ScopeRouter = "[TO BE CONFIRMED]"
$LeaseDuration = "[TO BE CONFIRMED]" # e.g. (New-TimeSpan -Days 8)

# --- DNS -----
$Forwarders = @"([TO BE CONFIRMED])"

# --- Misc -----
$TimeZoneId = "China Standard Time" # consistent with Shanghai/Beijing - CONFIRM
$NtpSource  = "[TO BE CONFIRMED]"

```

4.1 Common baseline — every Windows Server 2025 host

1. Boot the Windows Server 2025 ISO.
2. Select the edition per node: **Server Core** for the four DMZ servers (mandated by the topology); Core or Desktop Experience for INTERNAL hosts [TO BE CONFIRMED].
3. Set the local Administrator password from your credential vault. **Do not reuse a password across tiers.**
4. Apply the baseline below.

```

# Run on every Windows Server 2025 host as the first post-install action
Set-TimeZone -Id $TimeZoneId
Set-NetFirewallProfile -Profile Domain,Private,Public -Enabled True

# SMBv1 is not installed by default on Server 2025 - verify rather than remove

```

```
Get-WindowsOptionalFeature -Online -FeatureName SMB1Protocol |
    Select-Object FeatureName, State

# Enforce SMB signing and reject unencrypted guest access
Set-SmbServerConfiguration -RequireSecuritySignature $true -EnableSMB1Protocol $false -Force
Set-SmbClientConfiguration -RequireSecuritySignature $true -Force

# Confirm the OS build matches the 2025 baseline
Get-ComputerInfo -Property OsName, OsVersion, OsHardwareAbstractionLayer
```

Server Core hosts: there is no local Server Manager. Use `sconfig` for initial network and naming, then manage remotely via **Windows Admin Center** or Enter-PSSession. All GUI procedures below apply only to Desktop Experience nodes.

Network configuration pattern

```
# Apply per host, substituting the relevant variables
$if = (Get-NetAdapter | Where-Object Status -eq 'Up').Name

New-NetIPAddress -InterfaceAlias $if `
    -IPAddress $DC1IP `
    -PrefixLength [TO BE CONFIRMED] `
    -DefaultGateway $DC1Gateway

Set-DnsClientServerAddress -InterfaceAlias $if -ServerAddresses $DC1IP

Rename-Computer -NewName $DC1Name -Restart
```

4.2 First domain controller — forest root (primary site, INTERNAL zone)

Server Manager GUI path

1. **Manage > Add Roles and Features** > Role-based or feature-based installation.
2. Select **Active Directory Domain Services > Add Features > Install**.
3. Click the notification flag > **Promote this server to a domain controller**.
4. Select **Add a new forest**; enter the root domain name from `$ForestRoot`.
5. Set forest and domain functional levels per `$ForestMode` / `$DomainMode`.
6. Leave **DNS server** selected; set the DSRM password.
7. Accept the DNS delegation warning, review paths, **Install**, allow the restart.

PowerShell path

```
Install-WindowsFeature AD-Domain-Services,DNS -IncludeManagementTools

Import-Module ADDSDeployment
Install-ADDSForest `
    -DomainName $ForestRoot `
    -DomainNetbiosName $NetBiosName `
    -ForestMode $ForestMode `
    -DomainMode $DomainMode `
    -InstallDns:$true `
    -DatabasePath "C:\Windows\NTDS" -LogPath "C:\Windows\NTDS" -SysvolPath "C:\Windows\SYSVOL"
```

```
-SafeModeAdministratorPassword $Dsrmpassword `
-NoRebootOnCompletion:$false -Force
```

Post-promotion: authoritative time

```
# Run on the PDC Emulator only
w32tm /config /manualpeerlist:"$NtpSource" /syncfromflags:manual /reliable:yes /update
Restart-Service w32time
w32tm /resync /force
```

4.3 DNS configuration

```
# Reverse lookup zones - one per subnet, all eight
$AllSubnets = @($SH_DMZ_Subnet, $SH_INT_Subnet, $SH_CLI_Subnet, $SH_EDGE_Subnet,
                $BJ_DMZ_Subnet, $BJ_INT_Subnet, $BJ_CLI_Subnet, $BJ_EDGE_Subnet)

foreach ($net in $AllSubnets) {
    Add-DnsServerPrimaryZone -NetworkId $net -ReplicationScope Forest
}

# Forwarders
Set-DnsServerForwarder -IPAddress $Forwarders -UseRootHint $false

# Scavenging - prevents stale records in DHCP-served CLIENT zones
Set-DnsServerScavenging -ScavengingState $true -RefreshInterval 7.00:00:00 `
    -NoRefreshInterval 7.00:00:00 -ApplyOnAllZones
```

A and CNAME records for published DMZ services cannot be written here — the service names are redacted in the source. Add them once the values are released.

4.4 AD Sites & Services — before the second-site DC

```
# Create sites matching the physical topology
New-ADReplicationSite -Name $SitePrimary
New-ADReplicationSite -Name $SiteSecondary

# Associate subnets with sites
New-ADReplicationSubnet -Name $SH_DMZ_Subnet -Site $SitePrimary
New-ADReplicationSubnet -Name $SH_INT_Subnet -Site $SitePrimary
New-ADReplicationSubnet -Name $SH_CLI_Subnet -Site $SitePrimary
New-ADReplicationSubnet -Name $BJ_DMZ_Subnet -Site $SiteSecondary
New-ADReplicationSubnet -Name $BJ_INT_Subnet -Site $SiteSecondary
New-ADReplicationSubnet -Name $BJ_CLI_Subnet -Site $SiteSecondary

# Site link - cost and interval are [TO BE CONFIRMED]
New-ADReplicationSiteLink -Name "SH-BJ-Link" `
    -SitesIncluded $SitePrimary,$SiteSecondary `
    -Cost 100 -ReplicationFrequencyInMinutes 15 `
    -OtherAttributes @{'options'=1} # enable change notification

# Remove the default site link if it is no longer appropriate
# Remove-ADReplicationSiteLink -Identity "DEFAULTIPSITELINK" -Confirm:$false
```

GUI: dssite.msc > right-click **Sites** > New Site > name and select a link. Then Subnets > New Subnet > enter prefix and associate the site.

4.5 Additional domain controllers

Applies to the second DC in the primary site and the DC in the secondary site. **The secondary-site DC requires Phase 2 transport and Phase 5 site definitions to be complete.**

```
Install-WindowsFeature AD-Domain-Services,DNS -IncludeManagementTools

Import-Module ADDSDeployment
Install-ADDSDomainController `
    -DomainName                $ForestRoot `
    -SiteName                   $SiteSecondary `
    -InstallDns:$true `
    -Credential                (Get-Credential "$NetBiosName\Administrator") `
    -DatabasePath "C:\Windows\NTDS" -LogPath "C:\Windows\NTDS" -SysvolPath "C:\Windows\SYVOL" `
    -SafeModeAdministratorPassword $DsrMPassword `
    -NoGlobalCatalog:$false `
    -NoRebootOnCompletion:$false -Force
```

After reboot, repoint DNS so the DC resolves against a partner first and itself second:

```
$if = (Get-NetAdapter | Where-Object Status -eq 'Up').Name
Set-DnsClientServerAddress -InterfaceAlias $if -ServerAddresses $DC1IP, "127.0.0.1"
```

4.6 DHCP — CLIENT zones, both sites

```
Install-WindowsFeature DHCP -IncludeManagementTools
Add-DhcpServerInDC -DnsName "$env:COMPUTERNAME.$ForestRoot" -IPAddress [TO BE CONFIRMED]
Set-ItemProperty -Path HKLM:\SOFTWARE\Microsoft\ServerManager\Roles\12 -Name ConfigurationState -Value 2

# Shanghai CLIENT scope
Add-DhcpServerv4Scope -Name "$SitePrimary-CLIENT" `
    -StartRange $SH_ScopeStart -EndRange $SH_ScopeEnd `
    -SubnetMask $SH_ScopeMask -State Active
Set-DhcpServerv4OptionValue -ScopeId [TO BE CONFIRMED] `
    -Router $SH_ScopeRouter -DnsServer $DC1IP,$DC2IP -DnsDomain $ForestRoot

# Beijing CLIENT scope
Add-DhcpServerv4Scope -Name "$SiteSecondary-CLIENT" `
    -StartRange $BJ_ScopeStart -EndRange $BJ_ScopeEnd `
    -SubnetMask $BJ_ScopeMask -State Active
Set-DhcpServerv4OptionValue -ScopeId [TO BE CONFIRMED] `
    -Router $BJ_ScopeRouter -DnsServer $DC3IP,$DC1IP -DnsDomain $ForestRoot
```

Relay agent — required if the DHCP server is off-subnet from CLIENT

```
Install-WindowsFeature RemoteAccess,Routing -IncludeManagementTools
Install-RemoteAccess -VpnType RoutingOnly

netsh routing ip relay install
```

```
netsh routing ip relay add dhcpserver [TO BE CONFIRMED]
netsh routing ip relay add interface name="[TO BE CONFIRMED]"
netsh routing ip relay set interface name="[TO BE CONFIRMED]" relaymode=enable maxhop=4 minsecs=4
```

4.7 PKI — AD CS, INTERNAL zone only

Never place a certification authority in the DMZ. Publish enrolment endpoints through the DMZ if required; keep the CA itself in INTERNAL.

Recommended hierarchy

Given a DMZ-facing estate, a **two-tier** hierarchy is the appropriate design: an offline standalone root and an online enterprise issuing CA. The source specifies no hierarchy, so this is a recommendation — [TO BE CONFIRMED].

```
# --- Tier 1: offline standalone root CA (not domain-joined, powered off after issuance)
Install-WindowsFeature ADCS-Cert-Authority -IncludeManagementTools
Install-AdcsCertificationAuthority -CAType StandaloneRootCA `
  -CACommonName $CACommonName `
  -KeyLength 4096 -HashAlgorithmName SHA256 `
  -ValidityPeriod Years -ValidityPeriodUnits 20 -Force

# --- Tier 2: online enterprise subordinate issuing CA (domain-joined, INTERNAL)
Install-WindowsFeature ADCS-Cert-Authority,ADCS-Web-Enrollment -IncludeManagementTools
Install-AdcsCertificationAuthority -CAType EnterpriseSubordinateCA `
  -CACommonName "[TO BE CONFIRMED]" `
  -KeyLength 4096 -HashAlgorithmName SHA256 `
  -OutputCertRequestFile "C:\SubCA.req" -Force
# Submit C:\SubCA.req to the offline root, then install the issued certificate:
# certutil -installcert C:\SubCA.crt
Install-AdcsWebEnrollment -Force
```

Publish templates and enable autoenrolment

```
certutil -SetCATemplates +WebServer
certutil -SetCATemplates +Machine

# Autoenrolment GPO
New-GPO -Name "PKI-Autoenrollment" | New-GPLink -Target "[TO BE CONFIRMED - domain or OU DN]"
Set-GPRegistryValue -Name "PKI-Autoenrollment" `
  -Key "HKLM\Software\Policies\Microsoft\Cryptography\AutoEnrollment" `
  -ValueName "AEPolicy" -Type DWord -Value 7
```

GUI: certsrv.msc > Certificate Templates > New > Certificate Template to Issue. Then certtmpl.msc > template > Properties > Security > grant **Enroll** and **Autoenroll**.

4.8 DMZ Core servers — built last, published last

All four DMZ nodes are **Windows Server 2025 Core**. Every action is remote or command-line.

```
# Initial configuration via sconfig, or remotely:
Enter-PSSession -ComputerName [TO BE CONFIRMED] -Credential (Get-Credential)
```

```
# Web tier with reverse proxy
Install-WindowsFeature Web-Server,Web-Mgmt-Service
Start-Process msixexec.exe -Wait -ArgumentList '/i "C:\media\rewrite_amd64_en-US.msi" /qn'
Start-Process msixexec.exe -Wait -ArgumentList '/i "C:\media\requestRouter_amd64.msi" /qn'

Set-WebConfigurationProperty -PSPath 'MACHINE/WEBROOT/APPHOST' `
-Filter "system.webServer/proxy" -Name "enabled" -Value "True"
```

Domain join decision [TO BE CONFIRMED]. The source does not indicate whether DMZ servers are domain members, members of a separate resource forest, or workgroup hosts. Standard practice for an exposed tier is workgroup or a dedicated forest. Resolve before Phase 12.

Site bindings, host headers and origin addresses are all [TO BE CONFIRMED] — the service names they would reference are redacted in the source.

4.9 CLIENT endpoints — Windows 11

```
Rename-Computer -NewName "[TO BE CONFIRMED]" -Restart
Add-Computer -DomainName $ForestRoot -Credential (Get-Credential "$NetBiosName\Administrator") -Restart
```

RSAT — explicitly required by the 2026 topology

```
# Install the full RSAT suite
Get-WindowsCapability -Online -Name "Rsat.*" |
  Where-Object State -eq "NotPresent" |
  Add-WindowsCapability -Online

# Or install selectively
Add-WindowsCapability -Online -Name "Rsat.ActiveDirectory.DS-LDS.Tools~~~~0.0.1.0"
Add-WindowsCapability -Online -Name "Rsat.GroupPolicy.Management.Tools~~~~0.0.1.0"
Add-WindowsCapability -Online -Name "Rsat.Dns.Tools~~~~0.0.1.0"
Add-WindowsCapability -Online -Name "Rsat.DHCP.Tools~~~~0.0.1.0"
Add-WindowsCapability -Online -Name "Rsat.CertificateServices.Tools~~~~0.0.1.0"
Add-WindowsCapability -Online -Name "Rsat.FileServices.Tools~~~~0.0.1.0"
```

WSL with Debian — explicitly required by the 2026 topology

```
# Requires virtualization enabled in firmware
wsl --install --no-distribution
Restart-Computer

# After reboot
wsl --set-default-version 2
wsl --install -d Debian
wsl --list --verbose
```

Inside Debian, if the distribution will be used for domain-aware automation:

```
sudo apt update && sudo apt install -y krb5-user python3-pip
pip3 install pywinrm[kerberos] ansible
# /etc/krb5.conf realm = [TO BE CONFIRMED - uppercase forest root]
```

5. Configuration Details

Every value in this section is [TO BE CONFIRMED]. The structures below are the architecturally required shapes; populate them from the released design.

5.1 AD DS structure

Object	Value	Status
OU hierarchy	[TO BE CONFIRMED] — recommend site-tier-function, e.g. OU=Shanghai > OU=Servers / OU=Workstations / OU=Users	Not in source
Security groups	[TO BE CONFIRMED]	Not in source
Service accounts	[TO BE CONFIRMED] — use group Managed Service Accounts (gMSA) wherever the service supports them	Not in source
Delegation model	[TO BE CONFIRMED] — delegate at OU level; avoid Domain Admin for routine operations	Not in source
Tiered admin model	Recommended: Tier 0 (DCs, CA), Tier 1 (INTERNAL servers), Tier 2 (CLIENT) with separate credentials per tier	Not in source

```
# Structural pattern - replace all placeholder names
$OuRoot = "DC=[TBC],DC=[TBC]"

New-ADOrganizationalUnit -Name $SitePrimary -Path $OuRoot
New-ADOrganizationalUnit -Name $SiteSecondary -Path $OuRoot

foreach ($site in @($SitePrimary, $SiteSecondary)) {
    $sitePath = "OU=$site,$OuRoot"
    "Servers","Workstations","Users","Groups","ServiceAccounts" | ForEach-Object {
        New-ADOrganizationalUnit -Name $_ -Path $sitePath -ProtectedFromAccidentalDeletion $true
    }
}
```

5.2 Group Policy objects

The source specifies no policy settings. The table below lists the GPOs a four-zone Windows Server 2025 estate requires; all settings are [TO BE CONFIRMED].

GPO	Link target	Purpose
Baseline-Security-Servers	Servers OUs, both sites	Windows Server 2025 security baseline
Baseline-Security-Clients	Workstations OUs, both sites	Windows 11 security baseline

GPO	Link target	Purpose
PKI-Autoenrollment	Domain root	Certificate autoenrolment and root trust distribution
Tier0-Restrictions	Domain Controllers OU	Deny logon to Tier 1 and Tier 2 credentials
DMZ-Hardening	DMZ servers [TBC]	Attack-surface reduction for the exposed tier
Client-Config	Workstations OUs	Drive maps, RSAT deployment, WSL policy
Windows-Update	All server and client OUs	Patch rings and maintenance windows [TBC]
Audit-Policy	Domain root	Advanced audit policy for forwarding to monitoring [TBC]

Creation and linking pattern

```
New-GPO -Name "Baseline-Security-Servers" |
  New-GPLink -Target "OU=Servers,OU=$SitePrimary,$OuRoot" -LinkEnabled Yes
New-GPLink -Name "Baseline-Security-Servers" `
  -Target "OU=Servers,OU=$SiteSecondary,$OuRoot" -LinkEnabled Yes
```

5.3 DNS zones and forwarders

Server	Zone	Type	Notes
Primary-site DCs	[TO BE CONFIRMED]	AD-integrated primary	Forest-wide replication scope
All DCs	Eight reverse zones	AD-integrated primary	One per zone per site
All DCs	Forwarders	—	[TO BE CONFIRMED]
DMZ resolvers	[TO BE CONFIRMED]	[TO BE CONFIRMED]	Split-brain DNS is likely given the DMZ; confirm whether external names resolve differently internally

5.4 DHCP scopes, reservations and options

Site	Scope	Range	Exclusions	Lease	Options
Shanghai	CLIENT	[TBC]	[TBC]	[TBC]	003 Router [TBC], 006 DNS [TBC], 015 Domain [TBC]
Beijing	CLIENT	[TBC]	[TBC]	[TBC]	003 Router [TBC], 006 DNS [TBC], 015 Domain [TBC]

Reservations: [TO BE CONFIRMED] — none are indicated in the source.

Recommendation: configure DHCP failover in load-balance mode between the two site servers once both scopes exist.

```
Add-DhcpServerv4Failover -Name "SH-BJ-Failover" `
  -PartnerServer "[TO BE CONFIRMED]" `
  -ScopeId [TO BE CONFIRMED] `
  -LoadBalancePercent 50 -SharedSecret "[TO BE CONFIRMED]"
```

5.5 File shares and NTFS permissions

No shares, paths, quotas or permissions appear in the source. All [TO BE CONFIRMED].

Standard to apply when values arrive:

- Share permissions: Authenticated Users – Change; **never** Everyone – Full Control unless an automation requirement forces it.
- NTFS: break inheritance at the share root; grant SYSTEM and Administrators Full, and the owning security group Modify.
- Enable Access-Based Enumeration on all group shares.
- Apply FSRM quotas and file screens if a data-governance requirement exists [TBC].

5.6 Certificate templates

Template	Purpose	Enrolment permission	Status
Web Server	HTTPS bindings for DMZ-published and INTERNAL services	Domain Computers — Enroll	[TBC]
Computer	LDAPS on DCs; IPsec	Domain Computers — Enroll, Autoenroll	[TBC]
[TO BE CONFIRMED]	Any custom template the released design requires	[TBC]	[TBC]

5.7 Print, RDS and other services

The source contains **no evidence** of print services, RDS or RemoteApp requirements. Do not build them unless the released design introduces them. If RDS is added, note the RDS CAL obligation beyond the 120-day grace period.

6. Validation & Testing

These checks are value-independent and can be run as written once hostnames are known.

Phase	Command	Expected healthy output
-------	---------	-------------------------

Phase	Command	Expected healthy output
1 — Zone traversal	Test-NetConnection [DMZ host] -Port 443; Test-NetConnection [INTERNAL host] -Port 445; Test-NetConnection [CLIENT gateway] -TraceRoute	Permitted paths return TcpTestSucceeded : True; blocked paths must return False — verify the deny rules as rigorously as the allow rules
2 — Inter-site transport	Test-NetConnection [Beijing host] -TraceRoute; Get-VpnS2SInterface \\ f1 ConnectionState,Persistent	Route traverses the expected hops; Connected and Persistent True if VPN
3 — Forest	dcdiag /v /c /e; Get-ADForest \\ f1 Name,ForestMode,GlobalCatalogs; Get-ADDomain \\ f1 Name,DomainMode,PDCEmulator	All dcdiag tests passed; forest and domain modes match \$ForestMode / \$DomainMode
3 — SRV records	Resolve-DnsName _ldap._tcp.dc._msdcs.\$ForestRoot -Type SRV; nltest /dsgetdc:\$ForestRoot	SRV records return the expected DC set; nltest returns a DC in the correct site
3 — Time	w32tm /query /status; w32tm /query /source	Stratum below 5; source matches \$NtpSource; skew under 5 minutes estate-wide
4 — DNS	Get-DnsServerZone; Resolve-DnsName [host IP] -Type PTR; Get-DnsServerForwarder	Eight reverse zones present; PTR resolves; forwarders match \$Forwarders
5 — Sites	Get-ADReplicationSite -Filter *; Get-ADReplicationSubnet -Filter * \\ ft Name,Site; nltest /dsgetsite	Shanghai and Beijing exist; every subnet is mapped to a site — unmapped subnets are the most common cause of cross-WAN authentication
6/7 — Replication	repadmin /replsummary; repadmin /showrepl * /csv; repadmin /syncall /AdeP	Zero fails, zero errors; all naming contexts Successful; largest delta within the site-link interval
6/7 — DC health	dcdiag /test:replications /test:advertising /test:fsmocheck /test:dns	All tests passed; FSMO roles held by the expected DCs
8 — DHCP	Get-DhcpServerInDC; Get-DhcpServerv4Scope; Get-DhcpServerv4Statistics; Get-DhcpServerv4Failover	Servers authorised; both CLIENT scopes Active; addresses in use greater than zero after client build; failover Normal
8 — Relay	From a CLIENT endpoint: ipconfig /release; ipconfig /renew	Address issued from the correct site scope; ipconfig /all shows the expected DHCP server
9 — PKI	certutil -ping; certutil -verifystore -enterprise Root; certutil -verify -urlfetch [cert file]; Get-CACr1DistributionPoint	CA responds; root present in the enterprise store; CDP and AIA URLs resolve and fetch — the most common PKI failure
10 — Templates	certutil -CATemplates; on a client certutil -pulse then Get-ChildItem Cert:\LocalMachine\My	Templates listed as issuable; autoenrolled certificate appears without manual action
11 — Members	Test-ComputerSecureChannel -Verbose; nltest /sc_verify:\$ForestRoot	True; secure channel to a DC in the correct site

Phase	Command	Expected healthy output
11 — LDAPS	Test-NetConnection [DC] -Port 636; ldp.exe bind on 636 with SSL	Port open; LDP binds with no certificate error
12 — DMZ	Get-Website; Invoke-WebRequest https://[published FQDN]; from DMZ attempt Test-NetConnection [INTERNAL host] -Port 445	Sites running; HTTPS returns 200 with a valid chain; SMB from DMZ to INTERNAL must fail
14 — Group Policy	gpresult /r /scope:computer; gpresult /h report.html; Get-GPOReport -All -ReportType Html -Path all.html	Expected GPOs in Applied list; no Denied (Security) on intended targets; no inheritance conflicts
15 — Shares and DFS	Get-SmbShare; Get-SmbShareAccess; Get-DfsnRoot; Get-DfsrBacklog	Shares present with correct ACLs; namespace Online; backlog 0
16 — Clients	Get-WindowsCapability -Online -Name "Rsat.*" \ Where State -eq Installed; wsl --list --verbose	RSAT capabilities Installed; Debian listed at version 2 , state Running Or Stopped
16 — Client auth	nltest /dsgetdc:\$ForestRoot /site:[site]; klist	Client authenticates against its local site DC , not across the WAN; Kerberos tickets present, no NTLM fallback
17 — Publication	From an external host: Invoke-WebRequest https://[published FQDN]; nmap or port scan of the edge [TBC]	Published services reachable; all other ports closed
18 — Backup	Restore test of a DC System State to isolated media	Authoritative restore completes; ntdsutil snapshot mounts

6.1 Estate-wide acceptance gate

Run before handover. All must pass.

```
# 1. Replication health across the forest
repadmin /replsummary
repadmin /showrepl * /csv | ConvertFrom-Csv | Where-Object {$_. "Number of Failures" -gt 0}

# 2. DC diagnostics on every DC
Get-ADDomainController -Filter * | ForEach-Object {
    Write-Host "=== $($_.HostName) ===" -ForegroundColor Cyan
    dcdiag /s:$($_.HostName) /q
}

# 3. Every subnet mapped to a site
Get-ADReplicationSubnet -Filter * | Select-Object Name, Site |
    Where-Object { -not $_.Site }

# 4. Time skew across the estate
Get-ADComputer -Filter {OperatingSystem -like "*Server*"} |
```

```

ForEach-Object { w32tm /stripchart /computer:$($_.DNSHostName) /samples:1 /dataonly }

# 5. Certificate chain validation on every host holding a machine certificate
Get-ChildItem Cert:\LocalMachine\My | ForEach-Object {
    [PSCustomObject]@{
        Subject = $_.Subject
        Expires = $_.NotAfter
        Valid    = (Test-Certificate -Cert $_ -ErrorAction SilentlyContinue)
    }
}

```

7. Gaps, Conflicts & Risks

7.1 The overriding gap

Gap	Impact
The source is a pre-release with every FQDN, IP, subnet, domain name and service label redacted	A build-ready plan cannot be produced. This document is a framework; it is not executable until the unredacted topology is released. No other gap in this section matters until this one is closed.

7.2 Gaps in the 2026 source

#	Gap	Impact	Action
G1	No hostnames or FQDNs	No naming standard can be validated; no certificate subject names can be planned	Obtain the naming convention with the released topology
G2	No IP addressing, masks or VLAN IDs	Zone segmentation cannot be implemented or firewalled	Obtain the full addressing plan
G3	No domain or forest names	Cannot determine single-domain versus multi-domain, or trust requirements	Obtain the AD design document
G4	No service-to-host mapping — every service label is redacted	Section 2 role placement is inference, not specification	Obtain the service catalogue. This is the highest-value missing item after addressing
G5	No edge firewall product or rule set, despite a DMZ zone being central to the design	The defining feature of the architecture is unimplementable	Obtain the firewall design and rule matrix
G6	No storage, hypervisor or VM specifications	Cannot size, procure or provision	Obtain the platform design

#	Gap	Impact	Action
G7	No backup, antivirus/EDR or monitoring	Estate would go live unprotected and unobserved	Specify and procure independently; do not wait on the topology
G8	No inter-site link type between Shanghai and Beijing	Cannot plan replication interval, bandwidth or resilience	Confirm WAN, MPLS or VPN
G9	No PKI hierarchy specification	Two-tier design in section 4.7 is a recommendation only	Confirm hierarchy depth and CA naming
G10	DMZ server domain membership not stated	Determines whether DMZ hosts are domain members, a resource forest, or workgroup	Resolve before Phase 12 — it changes the build materially
G11	No DNS namespace strategy	Split-brain DNS is probable given a DMZ, but unconfirmed	Confirm internal and external namespace alignment
G12	INTERNAL zone node count and composition unknown	Cannot produce a true node inventory	Obtain the full VM list

7.3 Internal conflicts and ambiguities in the source

#	Issue	Detail	Resolution
A1	Core versus GUI selection is unresolved	The topology shows both Windows Server 2025 Core and Desktop Experience in scope but only firmly assigns Core to the DMZ tier	Confirm per-host edition. Default to Core for all infrastructure roles; reserve GUI for a single management host
A2	Two Core servers per DMZ, purpose unstated	Could be an HA pair, a split-role pair, or two unrelated services	Confirm. If HA, a load-balancing mechanism is required and is not in the source
A3	Symmetric four-zone stack at both sites, relationship unstated	Active/active, active/passive and independent operation produce very different builds	Confirm the operating model before Phase 5
A4	Windows 11 build not specified	Feature availability and baseline applicability vary by build	Confirm the target build and servicing channel

7.4 Design risks

Risk	Severity	Description	Recommendation
Building against a redacted source	Critical	Any value assumed now will likely	Do not begin Phase 3 or later until

Risk	Severity	Description	Recommendation
		conflict with the released design, forcing rework or, worse, silent misconfiguration	the unredacted topology is released. Phases 0 to 2 can proceed on the zone architecture alone
No backup capability specified	High	A two-site forest with no defined backup has no recovery path from AD corruption or ransomware	Specify, procure and test restores before go-live. Include AD System State and an isolated copy
No EDR specified, with an internet-facing DMZ	High	Four Core servers in a semi-trusted zone with no endpoint detection	Deploy EDR to DMZ hosts as the first agent installed, before publication
DMZ-to-INTERNAL boundary undefined	High	The DMZ exists precisely to constrain this path; without a rule set it is either fully open or fully blocked	Define an explicit, minimal, brokered rule set. Deny by default
Single CA tier if a one-tier PKI is chosen	Medium	An online root compromise invalidates the entire trust chain	Adopt the two-tier design in section 4.7; keep the root offline
Single DC per site	Medium	Not stated either way, but a two-DC forest across two sites means one DC per site	Deploy two DCs per site, or accept that a site loses local authentication on a single failure
Inter-site dependency on one link	Medium	If Shanghai and Beijing share a single path, its loss halts replication	Confirm redundancy; if none, ensure each site has a local DC, GC, DNS and DHCP so it survives isolation
No monitoring	Medium	Replication failures and certificate expiry are silent until they cause an outage	Deploy monitoring covering replication, DC health, certificate expiry, disk and DHCP pool depletion
WSL on CLIENT endpoints	Medium	WSL is mandated by the topology but introduces a Linux attack surface and a credential path into the domain	Govern via policy: restrict distributions, disable unless required, and treat WSL Kerberos tickets as domain credentials
No tiered administration model specified	Medium	Domain Admin credentials used on CLIENT endpoints would collapse the security boundary	Implement Tier 0/1/2 separation from day one; it is far harder to retrofit
Time synchronisation not specified	Low	Kerberos fails above five minutes of skew, particularly across sites	Configure the PDC Emulator against an authoritative external source; let the hierarchy inherit

8. Pre-Implementation Checklist

8.1 Blocking prerequisite

- ☐ **Released, unredacted version of wsc2026_TP39_topology_pre.pdf obtained.** Nothing from Phase 3 onward can be built without it. This is the single blocking item.

8.2 Design documents to obtain

- ☐ Hostname and FQDN naming convention
- ☐ Full IP addressing plan with masks and VLAN IDs for all eight zone/site segments
- ☐ Forest and domain design, including functional levels and any trusts
- ☐ Service catalogue: which role runs on which host
- ☐ Edge firewall design and the Internet / DMZ / INTERNAL / CLIENT rule matrix
- ☐ Inter-site link specification: type, bandwidth, redundancy
- ☐ PKI design: hierarchy depth, CA naming, template requirements
- ☐ DNS namespace strategy, including split-brain resolution if applicable
- ☐ DMZ server domain-membership decision
- ☐ Storage and virtualization platform design

8.3 Hardware and platform

- ☐ Hypervisor hosts sized for the full VM count [TO BE CONFIRMED – inventory not in source]
- ☐ Virtual switches and VLANs for eight segments minimum: DMZ, INTERNAL, CLIENT and edge transit at each of Shanghai and Beijing
- ☐ Edge firewall appliances or virtual appliances at both sites
- ☐ Multi-NIC provisioning for every gateway and DMZ host
- ☐ Storage capacity for the server estate plus backup retention [TO BE CONFIRMED]
- ☐ Out-of-band management access to both sites

8.4 Media

- ☐ Windows Server 2025 ISO — both Server Core and Desktop Experience install options
- ☐ Windows 11 ISO at the confirmed build [TO BE CONFIRMED]
- ☐ Debian image for WSL, or confirmed internet access for `wsl --install -d Debian`
- ☐ IIS URL Rewrite (`rewrite_amd64_en-US.msi`) and Application Request Routing (`requestRouter_amd64.msi`) staged locally for the Core DMZ servers
- ☐ Windows Admin Center installer — required to manage four GUI-less DMZ servers
- ☐ Current cumulative updates for Server 2025 and Windows 11, staged offline

8.5 Licensing

- ☐ Windows Server 2025 licences — Standard or Datacenter per VM density [TO BE CONFIRMED]

- ☐ Windows Server CALs for the confirmed user count [TO BE CONFIRMED]
- ☐ Windows 11 licences for all CLIENT endpoints [TO BE CONFIRMED]
- ☐ RDS CALs **only if** RDS is confirmed in scope — not evidenced in the source
- ☐ Backup, EDR and monitoring licences [TO BE CONFIRMED]

8.6 Credentials and accounts

- ☐ Local Administrator passwords, **unique per host**, held in a credential vault
- ☐ Windows LAPS configured for all local Administrator accounts
- ☐ DSRM password per DC, recorded and tested
- ☐ Enterprise Admin account for forest creation and AD CS installation
- ☐ Tier 0, Tier 1 and Tier 2 administrative accounts, separated from day one
- ☐ Break-glass account created, documented and tested **before** any restrictive GPO is linked
- ☐ gMSA accounts identified for services that support them
- ☐ Firewall and hypervisor administrative credentials

8.7 Change approvals

- ☐ Approval to create the forest and its domain structure
- ☐ Approval for the inter-site link and its security model
- ☐ Approval for the edge firewall rule set, signed off by security
- ☐ Approval for the PKI hierarchy and CA naming, which is effectively permanent
- ☐ Approval for the DMZ publication model and the services to be exposed
- ☐ Approval for tiered administration and the associated operational change
- ☐ Maintenance windows for Phases 0 to 18, with snapshot and rollback points at Phases 3, 7, 9 and 12
- ☐ Security review scheduled between Phase 12 and Phase 17 — **before anything is published to the Internet zone**